

Microservices Lab

Assignment Week – 03

23-09-2026

Assignment: Book Management REST API

Assignment:

Objective: Build a Spring Boot + MySQL CRUD application for managing books, and test every endpoint using Postman.

Part 1 – Project Setup

1. Create a new Spring Boot project (spring-initializr or your IDE) with dependencies: Spring Web, Spring Data JPA, MySQL Driver.
2. Configure application.properties to connect to a MySQL database named bookstore_db.

Part 2 – The Entity

Create a Book entity with the following fields:

- id (Long, auto-generated primary key)
- title (String)
- author (String)
- publisher (String)
- price (double)
- publicationYear (int)

Use the correct JPA annotations (@Entity, @Id, @GeneratedValue) and generate/write getters and setters.

Part 3 – The Repository

Create BookRepository extending JpaRepository<Book, Long>.

Add **three custom finder methods**:

1. A method-name query to find all books by a given author.
2. A method-name query to find all books by a given publisher.
3. A @Query-based method to find all books with price between two given values (min and max).

Think about which of the three should use plain method-name matching, and which one benefits from @Query instead.

Part 4 – The Controller

Create BookController with the following endpoints:

Method	URL	Action
GET	/books	Get all books
GET	/books/{id}	Get a book by id
POST	/books	Create a new book
PUT	/books/{id}	Update an existing book
DELETE	/books/{id}	Delete a book
GET	/books/author/{author}	Get all books by a given author
GET	/books/publisher/{publisher}	Get all books by a given publisher
GET	/books/price-range?min=X&max=Y	Get all books within a price range

Part 5 – Testing with Postman

For each endpoint above:

1. Create the request in Postman with the correct method and URL.
 2. For POST and PUT, use Body → raw → JSON with sample data.
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